

Multiple Choice (10 marks)

1. Simplify and write the result with a rational denominator: $\sqrt{\sqrt[3]{\sqrt{\frac{1}{729}}}}$

(a) $\overline{\{3\sqrt{3}\}\{3\}}$

1. $\frac{1}{3}$

2. $\sqrt{3}$

3. $\overline{\{\sqrt{3}\}\{3\}}$

If there is a 70% chance that it will rain on a given day, what is the percent chance that it will NOT rain on that same day?

(a) 0%

(b) 30%

(c) 50%

(d) 70%

What is three fifth of 100?

(a) 3

(b) 5

(c) 20

(d) 60

The area bounded by the parabola $y = x^2$ and the lines $y = 1$ and $y = 9$ equals

(a) 8

(b) $84/3$

(c) $64\sqrt{2}/3$

(d) $104/3$

When $n = 11$, what is the value of $10 - (n + 6)$?

(a) -7

(b) 5

(c) 7

(d) 27

True / False (10 marks)

Answer True or False

6. True or False: The relative maximum value of the function $y = (\ln x)/x$ occurs at $x = e$, yielding a maximum value of $1/e$.
7. True or False: The quotient of 1,248 divided by 7 is 178 with a remainder of 2.
8. True or False: The point $(5, 5)$ can lie on the circle that passes through the points $(3, 4)$ and $(5, 7)$.
9. True or False: Noah rides his bike at a speed of 10 miles per hour.
10. True or False: The P-value for the hypothesis test $H_0: p=0.3$ against $H_A: p>0.3$ is 1.06.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In an algebra test with 35 problems, you need to score at least 80% to pass. Analyze the number of problems you can miss while still achieving a passing score, and explain your reasoning.
12. Calculate the area of a square whose perimeter equals that of an equilateral triangle with a side length of 8. Explain your reasoning and the steps involved in your calculation.

End of Paper